

Problem Set 3

1. Warm up: In S_4 , show that $s_1 s_2 s_1 \leq s_2 s_1 s_3 s_2$ and that the elements $s_1 s_2 s_3$ and $s_2 s_1 s_3 s_2$ are incomparable in Bruhat order.
2. Prove that there is a reflection $t \in T$ such that $u = tv$ if and only if there is a reflection $t' \in T$ such that $u = vt'$.
3. Prove, for $w \in W$, that the total number of incoming edges to and outgoing edges from w in the Bruhat graph is $|T|$.
4. How many total edges are there in the Bruhat graph of S_n ? How many total cover relations are in the weak order on S_n ?
5. Fix $1 \leq k < n$. A permutation is called **k -Grassmannian** if it is equal to e or has s_k as its unique descent.
 - (a) Show that taking the set $X_\pi = \{\pi(1), \dots, \pi(k)\}$ gives a bijection between k -Grassmannian permutations and subsets of $[n]$ of size k .
 - (b) Show that k -Grassmannian permutations π_1, π_2 satisfy $\pi_1 \leq \pi_2$ if and only if $X_{\pi_1} \leq X_{\pi_2}$ in Gale order.
 - (c) Find a bijection between subsets of $[n]$ of size k and partitions that fit in a box of size $k \times (n - k)$. Show that $X \leq X'$ in Gale order if and only if the corresponding partitions λ, λ' satisfy $\lambda_i \leq \lambda'_i$ for all $1 \leq i \leq k$.
 - (d) Bonus: show that there is an analog of the Bruhat decomposition which says

$$\mathrm{GL}_n(\mathbb{C}) = \bigsqcup_{w \text{ is } k\text{-Grassmannian}} BwP_{n,k},$$

where $P_{n,k}$ is the group of matrices of the form $\begin{pmatrix} A & D \\ \mathbf{0} & C \end{pmatrix}$, where $A \in \mathrm{GL}_k(\mathbb{C})$ and $C \in \mathrm{GL}_{n-k}(\mathbb{C})$.

You saw in the Schubert calculus problem set that the Grassmannian $\mathrm{Gr}(k, n)$ is isomorphic to $\mathrm{GL}_n(\mathbb{C})/P_{n,k}$. Define the Schubert cell Ω'_λ to be $BwP_{n,k}/P_{n,k}$, where w is the k -Grassmannian permutation associated to λ . (Note this is not quite the same definition as in the Schubert calculus lectures; the Ω_λ defined there is isomorphic to our Ω'_{λ^c} .) It follows from what you have shown in this problem (and the compactness of the flag variety) that $\Omega'_{\lambda_1} \subseteq \overline{\Omega'_{\lambda_2}}$ if and only if $\lambda_1 \leq \lambda_2$.

6. Show that the length generating function of S_n

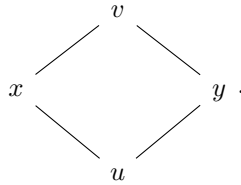
$$f(q) = \sum_{w \in S_n} q^{\ell(w)}$$

factors as

$$f(q) = (1+q)(1+q+q^2)(1+q+q^2+q^3) \cdots (1+q+\cdots+q^{n-1}).$$

The polynomial $1+q+\cdots+q^{k-1}$ is called a **q -integer** or **quantum integer**, denoted $[k]_q$. (This is because when you set $q=1$ you get the classical integer k .) The function $f(q)$ is called the **q -factorial**, denoted $[n]_q!$.

7. Prove that if $u \leq v$ and $\ell(v) = \ell(u) + 2$, then there are exactly two elements x such that $u < x < v$. Another way of phrasing this is to say that the **Bruhat interval** $[u, v]$ has the Hasse diagram



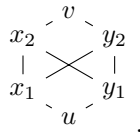
8. Prove that, in the symmetric group, $u \leq v$ if and only if $w_0 u w_0 \leq w_0 v w_0$ (this is in fact true for all finite Coxeter groups).
9. In the symmetric group, prove the **Tableau criterion**. This says that, to check that $u \leq v$, we just need to check that $\{u(1), \dots, u(k)\} \leq \{v(1), \dots, v(k)\}$ in Gale order for every k such that s_k is a descent of u .

10. In the symmetric group S_n , let's list all the transpositions as $t_1, t_2, \dots, t_{\binom{n}{2}}$ in lexicographic order, so that $t_1 = (1, 2), t_2 = (1, 3), \dots, t_{n-1} = (1, n), t_n = (2, 3), \dots, t_{\binom{n}{2}} = (n-1, n)$. Let $u < v$ be two permutations related by Bruhat order. Define a sequence of permutations by the following algorithm: set $u_0 = u$. At step i , do the following:

- If $u_{i-1} < t_i u_{i-1} \leq v$, then set $u_i = t_i u_{i-1}$.
- Otherwise, set $u_i = u_{i-1}$.

Prove that $u_{\binom{n}{2}} = v$. Moreover, show for all i that either $u_i = u_{i-1}$ or $\ell(u_i) = \ell(u_{i-1}) + 1$.

11. Assume $u \rightarrow v$ is an edge of the Bruhat graph such that $\ell(v) = \ell(u) + 3$. Prove that the Bruhat interval $[u, v] := \{x \in W \mid u \leq x \leq v\}$ has 6 elements, with Hasse diagram



Bonus: Prove the converse.

12. For each $w \in S_n$, show that there is an $\ell(w)$ -dimensional subgroup U_w of B such that the map $\phi : U_w \rightarrow \text{GL}_n(\mathbb{C})/B$ sending $u \in U_w$ to the coset uwB is bijective. (A consequence is that the Schubert cell BwB/B in the flag variety has complex dimension $\ell(w)$.)

13. The **Mobius function** of a poset P assigns to each pair $x \leq y$ an integer defined by

$$\mu(x, y) := \begin{cases} 1 & \text{if } x = y \\ -\sum_{x \leq z < y} \mu(x, z) & \text{if } x < y \end{cases}$$

Show that Bruhat order on W is an **Eulerian poset**, meaning $\mu(u, v) = (-1)^{\ell(v) - \ell(u)}$ for all $u \leq v$.